



Lecture 5: Floating point

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5 June 2026

Floating point numbers

Numbers

- So far, we only dealt with integers
- But there are other types of numbers



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- But there are other types of numbers
- Rational numbers (from ratio $\simeq$ fraction)
 - $3/4 = 0.75$
 - $10/3 = 3.33333333....$



Numbers

- So far, we only dealt with integers
- But there are other types of numbers
- Rational numbers (from ratio $\simeq$ fraction)
 - $3/4 = 0.75$
 - $10/3 = 3.33333333....$
- Real numbers
 - $\pi = 3.14159265...$
 - $e = 2.71828182...$



Very Large Numbers

- Distance of sun and earth

150,000,000,000 meters

- Scientific notation

1.5×10^{11} meters

- Another example: number of atoms in 12 gram of carbon-12 (1 mol)

$6.022140857 \times 10^{23}$



Binary Numbers in Scientific Notation

- Example binary number (π again)

11.0010010001

- Scientific notation

1.10010010001×2^1

- General form

$1.x \times 2^y$



Representation

- IEEE 754 floating point standard
- Uses 4 bytes

31	30	29	...	24	23	22	21	...	1	0
s	exponent					fraction				
1 bit	8 bits					23 bits				

- Exponent is offset with a bias of 127
e.g. $2^{-6} \rightarrow \text{exponent} = -6 + 127 = 121$

Conversion into Binary

- $\pi = 3.14159265$
- Number before period: $3_{10} = 11_2$
- Conversion of fraction .14159265



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	$0.14159265 \times 2 \downarrow$



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	$0.14159265 \times 2 \downarrow$
0	0.2831853



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	$0.14159265 \times 2 \downarrow$
0	$0.2831853 \times 2 \downarrow$
0	0.5663706



Conversion into Binary

- $\pi = 3.14159265$
- Number before period: $3_{10} = 11_2$
- Conversion of fraction .14159265

Digit	Calculation
	$0.14159265 \times 2 \downarrow$
0	$0.2831853 \times 2 \downarrow$
0	$0.5663706 \times 2 \downarrow$
1	0.1327412



Conversion into Binary

- $\pi = 3.14159265$
- Number before period: $3_{10} = 11_2$
- Conversion of fraction .14159265

Digit	Calculation	Digit	Calculation
	$0.14159265 \times 2 \downarrow$	1	$0.9817472 \times 2 \downarrow$
0	$0.2831853 \times 2 \downarrow$	1	$0.9634944 \times 2 \downarrow$
0	$0.5663706 \times 2 \downarrow$	1	$0.9269888 \times 2 \downarrow$
1	$0.1327412 \times 2 \downarrow$	1	$0.8539776 \times 2 \downarrow$
0	$0.2654824 \times 2 \downarrow$	1	$0.7079552 \times 2 \downarrow$
0	$0.5309648 \times 2 \downarrow$	1	$0.4159104 \times 2 \downarrow$
1	$0.0619296 \times 2 \downarrow$	0	$0.8318208 \times 2 \downarrow$
0	$0.1238592 \times 2 \downarrow$	1	$0.6636416 \times 2 \downarrow$
0	$0.2477184 \times 2 \downarrow$	1	$0.3272832 \times 2 \downarrow$
0	$0.4954368 \times 2 \downarrow$	0	$0.6545664 \times 2 \downarrow$
0	$0.9908736 \times 2 \rightarrow$	1	0.3091328×2

- Binary: 11.00100100001111101101



Encoding into Representation

- π

$$1.1001001000011111101101 \times 2^1$$

- Encoding

Sign	Exponent	Fraction
0	10000000	1001001000011111101101

- Note: leading 1 in fraction is omitted

Clicker quiz!

Clicker quiz omitted from public slides



See the representation of a float

```
#include <stdio.h>

int main(void) {
    float x;
    scanf("%f", &x);
    unsigned *p = (unsigned *) &x;

    for (int i = 31; i >= 0; i--) {
        printf("%c", (*p & (1 << i)) ? '1' : '0');
        if (i == 31 || i == 23) { printf(" "); }
    }
    printf("\n");

    return 0;
}
```



See the representation of a float

```
$ gcc explain.c  
$ echo '-18.8203125' | ./a.out  
1 10000011 001011010010000000000000
```



Special Cases

- Zero



Special Cases

- Zero
- Infinity ($1/0$)
- Negative infinity ($-1/0$)



Special Cases

- Zero
- Infinity ($1/0$)
- Negative infinity ($-1/0$)
- Not a number ($0/0$ or $\infty - \infty$)



Exponent	Fraction	Object
0	0	zero
0	>0	denormalized number
1-254	anything	floating point number
255	0	infinity
255	>0	NaN (not a number)

(denormalized number: $0.x \times 2^{-126}$)

Clicker quiz!

Clicker quiz omitted from public slides



Double Precision

- Single precision = 4 bytes

Sign	Exponent	Fraction
1 bit	8 bits	23 bits

- Double precision = 8 bytes

Sign	Exponent	Fraction
1 bit	11 bits	52 bits



Addition

Addition with Scientific Notation

- Decimal example, with 4 significant digits in encoding
- Example

$$0.1610 + 99.99$$

- In scientific notation

$$1.610 \times 10^{-1} + 9.999 \times 10^1$$



Addition with Scientific Notation

- Decimal example, with 4 significant digits in encoding
- Example

$$0.1610 + 99.99$$

- In scientific notation

$$1.610 \times 10^{-1} + 9.999 \times 10^1$$

- Bring lower number on same exponent as higher number

$$0.01610 \times 10^1 + 9.999 \times 10^1$$



Addition with Scientific Notation

- Round to 4 significant digits

$$0.016 \times 10^1 + 9.999 \times 10^1$$



Addition with Scientific Notation

- Round to 4 significant digits

$$0.016 \times 10^1 + 9.999 \times 10^1$$

- Add fractions

$$0.016 + 9.999 = 10.015$$



Addition with Scientific Notation

- Round to 4 significant digits

$$0.016 \times 10^1 + 9.999 \times 10^1$$

- Add fractions

$$0.016 + 9.999 = 10.015$$

- Adjust exponent

$$10.015 \times 10^1 = 1.0015 \times 10^2$$



Addition with Scientific Notation

- Round to 4 significant digits

$$0.016 \times 10^1 + 9.999 \times 10^1$$

- Add fractions

$$0.016 + 9.999 = 10.015$$

- Adjust exponent

$$10.015 \times 10^1 = 1.0015 \times 10^2$$

- Round to 4 significant digits

$$1.002 \times 10^2$$



Binary Floating Point Addition

- Numbers

$$0.5_{10} = \frac{1}{2}_{10}$$



Binary Floating Point Addition

- Numbers

$$0.5_{10} = \frac{1}{2}_{10} = \frac{1}{2^1}_{10}$$



Binary Floating Point Addition

- Numbers

$$0.5_{10} = \frac{1}{2}_{10} = \frac{1}{2^1}_{10} = 0.1_2$$



Binary Floating Point Addition

- Numbers

$$0.5_{10} = \frac{1}{2}_{10} = \frac{1}{2^1}_{10} = 0.1_2 = 1.000_2 \times 2^{-1}$$



Binary Floating Point Addition

- Numbers

$$0.5_{10} = \frac{1}{2}_{10} = \frac{1}{2^1}_{10} = 0.1_2 = 1.000_2 \times 2^{-1}$$

$$-0.4375_{10} = -\frac{7}{16}_{10}$$



Binary Floating Point Addition

- Numbers

$$0.5_{10} = \frac{1}{2}_{10} = \frac{1}{2^1}_{10} = 0.1_2 = 1.000_2 \times 2^{-1}$$

$$-0.4375_{10} = -\frac{7}{16}_{10} = -\frac{7}{2^4}_{10} = 0.0111_2$$



Binary Floating Point Addition

- Numbers

$$0.5_{10} = \frac{1}{2}_{10} = \frac{1}{2^1}_{10} = 0.1_2 = 1.000_2 \times 2^{-1}$$

$$-0.4375_{10} = -\frac{7}{16}_{10} = -\frac{7}{2^4}_{10} = 0.0111_2 = -1.110_2 \times 2^{-2}$$



Binary Floating Point Addition

- Numbers

$$0.5_{10} = \frac{1}{2}_{10} = \frac{1}{2^1}_{10} = 0.1_2 = 1.000_2 \times 2^{-1}$$

$$-0.4375_{10} = -\frac{7}{16}_{10} = -\frac{7}{2^4}_{10} = 0.0111_2 = -1.110_2 \times 2^{-2}$$

- Bring lower number on same exponent as higher number

$$-1.110 \times 2^{-2} = -0.111 \times 2^{-1}$$



Binary Floating Point Addition

- Numbers

$$0.5_{10} = \frac{1}{2}_{10} = \frac{1}{2^1}_{10} = 0.1_2 = 1.000_2 \times 2^{-1}$$

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- Bring lower number on same exponent as higher number

$$-1.110 \times 2^{-2} = -0.111 \times 2^{-1}$$

- Add the fractions

$$1.000_2 \times 2^{-1} + (-0.111 \times 2^{-1}) = 0.001 \times 2^{-1}$$



Binary Floating Point Addition

- Numbers

$$0.5_{10} = \frac{1}{2}_{10} = \frac{1}{2^1}_{10} = 0.1_2 = 1.000_2 \times 2^{-1}$$

$$-0.4375_{10} = -\frac{7}{16}_{10} = -\frac{7}{2^4}_{10} = 0.0111_2 = -1.110_2 \times 2^{-2}$$

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- Add the fractions

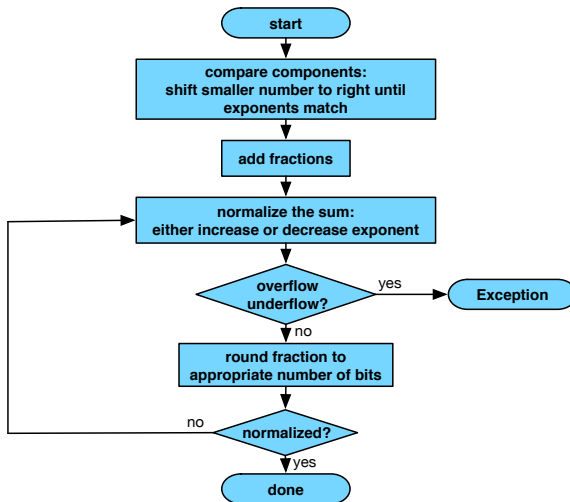
$$1.000_2 \times 2^{-1} + (-0.111 \times 2^{-1}) = 0.001 \times 2^{-1}$$

- Adjust exponent

$$0.001 \times 2^{-1} = 1.000 \times 2^{-4}$$



Flowchart



Multiplication



Multiplication with Scientific Notation

- Example: multiply 1.110×10^{10} and 9.200×10^{-5}



Multiplication with Scientific Notation

- Example: multiply 1.110×10^{10} and 9.200×10^{-5}
 $1.110 \times 10^{10} \times 9.200 \times 10^{-5}$



Multiplication with Scientific Notation

- Example: multiply 1.110×10^{10} and 9.200×10^{-5}

$$1.110 \times 10^{10} \times 9.200 \times 10^{-5}$$

$$1.110 \times 9.200 \times 10^{-5} \times 10^{10}$$



Multiplication with Scientific Notation

- Example: multiply 1.110×10^{10} and 9.200×10^{-5}

$$1.110 \times 10^{10} \times 9.200 \times 10^{-5}$$

$$1.110 \times 9.200 \times 10^{-5} \times 10^{10}$$

$$1.110 \times 9.200 \times 10^{-5+10}$$



Multiplication with Scientific Notation

- Example: multiply 1.110×10^{10} and 9.200×10^{-5}

$$1.110 \times 10^{10} \times 9.200 \times 10^{-5}$$

$$1.110 \times 9.200 \times 10^{-5} \times 10^{10}$$

$$1.110 \times 9.200 \times 10^{-5+10}$$

- Add exponents

$$-5 + 10 = 5$$



Multiplication with Scientific Notation

- Example: multiply 1.110×10^{10} and 9.200×10^{-5}

$$1.110 \times 10^{10} \times 9.200 \times 10^{-5}$$

$$1.110 \times 9.200 \times 10^{-5} \times 10^{10}$$

$$1.110 \times 9.200 \times 10^{-5+10}$$

- Add exponents

$$-5 + 10 = 5$$

- Multiply fractions

$$1.110 \times 9.200 = 10.212$$



Multiplication with Scientific Notation

- Example: multiply 1.110×10^{10} and 9.200×10^{-5}

$$1.110 \times 10^{10} \times 9.200 \times 10^{-5}$$

$$1.110 \times 9.200 \times 10^{-5} \times 10^{10}$$

$$1.110 \times 9.200 \times 10^{-5+10}$$

- Add exponents

$$-5 + 10 = 5$$

- Multiply fractions

$$1.110 \times 9.200 = 10.212$$

- Adjust exponent

$$10.212 \times 10^5 = 1.0212 \times 10^6$$

Binary Floating Point Multiplication

- Example

$$1.000 \times 2^{-1} \times -1.110 \times 2^{-2}$$



Binary Floating Point Multiplication

- Example

$$1.000 \times 2^{-1} \times -1.110 \times 2^{-2}$$

- Add exponents

$$-1 + (-2) = -3$$



Binary Floating Point Multiplication

- Example

$$1.000 \times 2^{-1} \times -1.110 \times 2^{-2}$$

- Add exponents

$$-1 + (-2) = -3$$

- Multiply fractions

$$1.000 \times -1.110 = -1.110$$



Binary Floating Point Multiplication

- Example

$$1.000 \times 2^{-1} \times -1.110 \times 2^{-2}$$

- Add exponents

$$-1 + (-2) = -3$$

- Multiply fractions

$$1.000 \times -1.110 = -1.110$$

$$1000 \times 1110 = 1110000$$



Binary Floating Point Multiplication

- Example

$$1.000 \times 2^{-1} \times -1.110 \times 2^{-2}$$

- Add exponents

$$-1 + (-2) = -3$$

- Multiply fractions

$$1.000 \times -1.110 = -1.110$$

$$1000 \times 1110 = 1110000$$

$$-1.110000$$



Binary Floating Point Multiplication

- Example

$$1.000 \times 2^{-1} \times -1.110 \times 2^{-2}$$

- Add exponents

$$-1 + (-2) = -3$$

- Multiply fractions

$$1.000 \times -1.110 = -1.110$$

$$1000 \times 1110 = 1110000$$

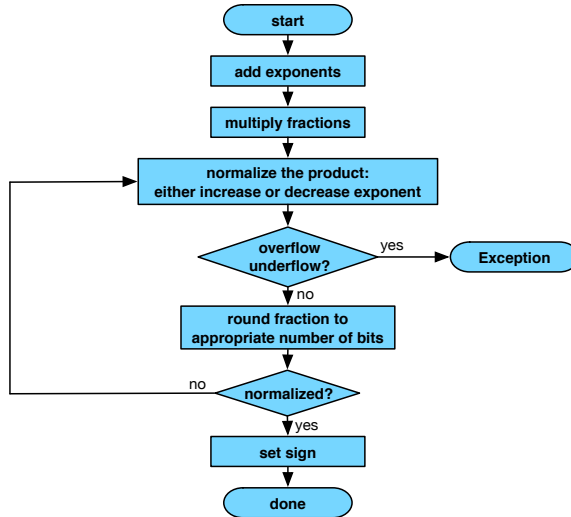
$$-1.110000$$

- Adjust exponent (not needed)

$$-1.110 \times 2^{-3}$$



Flowchart



Acknowledgements

Slides adapted from materials provided by David Hovemeyer.

